6.10.2026

Assume $A \in \mathbb{R}^{n \times n}$ square matrix, denote A_{ij} the $(n-1) \times (n-1)$ matrix obtained by deleting row i, column j

The determinant of A can be computed by Laplace expansion (along a row i)

$$\det(A) = \sum_{j=1}^{n} (-1)^{i+j} a_{ij} \det(A_{ij})$$

A square matrix A is called invertible if there exists A^{-1} ("inverse") such that

$$A^{-1} A = A A^{-1} = I_n$$

Following statements are equivalent:

1) A is invertible
2) $\operatorname{rank}(A) = n$
3) $\det(A) \neq 0$
4) The rows of A are linearly independent (l.i.)
5) The columns — " — ———
6) For every $b \in \mathbb{R}^n$, the system $Ax = b$ has exactly one solution

If any of conditions 1) - 6) hold, then solution to $Ax = b$ is

$$x = A^{-1} b.$$

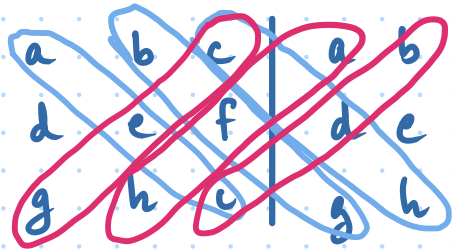
Example 1.8

2-D $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \rightsquigarrow \det(A) = a_{11}a_{22} - a_{12}a_{21}$

A invertible $\Leftrightarrow \det(A) \neq 0$

$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix}$ (Easy formula for inverse of 2x2 matrix)

3-D $A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \rightsquigarrow \det(A) = aei + bfg + cdh - ceg - afh - bdi$



rule of Sarrus
(Easy formula for determinant of 3x3 matrix)

Recall $Ax = b$ is solvable $\Leftrightarrow b \in \mathrm{col}(A)$

$\Leftrightarrow \mathrm{rank}(A) = \mathrm{rank}(A|b)$

If system solvable, then

-) it has a unique solution if $\text{rank}(A) = n$
-) it has infinitely many solutions if $\text{rank}(A) < n$

Example 1.9 (Unique, ∞-many, no solutions)

Consider

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 5 \\ 3 \end{pmatrix} \rightsquigarrow \det(A) = 1$$

$$A^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \rightsquigarrow x = A^{-1}b = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

} Unique soln

Consider

$$B = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \rightsquigarrow \det(B) = 0$$

$$Bx = \begin{pmatrix} 3 \\ 6 \end{pmatrix} \rightsquigarrow \text{infinitely many solutions}$$

or $Bx = \begin{pmatrix} 3 \\ 7 \end{pmatrix} \rightsquigarrow$ no solution

1.4 Linear optimization terminology

1.4.1 The standard form

Consider an example:

Example 1.10 (Matrix-vector notation for LPs)

$$\begin{aligned} \min \quad & 3x_1 + 2x_2 \\ \text{s.t.} \quad & x_1 + x_2 \leq 4 \\ & 2x_1 + x_2 \leq 5 \\ & x_1 + 3x_2 \leq 6 \\ & x \geq 0 \end{aligned} \qquad (\Leftrightarrow x_1 \geq 0,\ x_2 \geq 0)$$

Define

$$x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2, \quad c = \begin{pmatrix} 3 \\ 2 \end{pmatrix} \in \mathbb{R}^2, \quad b = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix} \in \mathbb{R}^3 \quad A = \begin{pmatrix} 1 & 1 \\ 2 & 1 \\ 1 & 3 \end{pmatrix} \in \mathbb{R}^{3\times 2}$$

Compact formulation:

$$\begin{aligned} \min \quad & c^T x \\ \text{s.t.} \quad & Ax \leq b \\ & x \geq 0 \end{aligned}$$

Definition 1.11

Let $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$. The problem

$$\begin{aligned} \max \quad & c^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0 \end{aligned}$$

is called an **LP in standard form**. The function $f(x) := c^T x$ is called the **objective function**. The expressions **$Ax = b$, $x \geq 0$** are called constraints on the **decision variable** x.

It will always be possible to transform an LP into a standard form. To see this, we also define

Definition 1.12

A decision variable x_i is called **free** if it has no sign restriction. (In the formulation of an LP, sometimes this is emphasized by writing "x_i free")

A **slack variable** is a nonnegative variable introduced to convert "$\leq$" into "$=$"

In particular, the scalar constraint

$$\alpha^T x = \alpha_1 x_1 + \dots + \alpha_n x_n \leq \beta$$

is equivalent to the two scalar constraints

$$\alpha^T x + s = \beta, \quad s \geq 0 \quad \rightsquigarrow \quad s = \beta - \alpha^T x \text{ represents the "slack" in the orginal inequality.}$$

Theorem 1.13

Every LP can be transformed into standard form.

Proof

- We have $\min(f) = -\max(-f)$, so any minimization problem can be written as a maximization problem.
- Any "$\leq$" can be transformed into "$=$" constraint by introducing a slack variable $s \geq 0$ (as in Def. 1.12)
- An inequality of the form $\alpha^T x \geq \beta \iff -\alpha^T x \leq -\beta$.
- A free variable x_i can be transformed by introducing two new variables x_i^+, x_i^- and setting

$$x_i = x_i^+ - x_i^-, \quad x_i^+, x_i^- \geq 0$$

Example 1.14 (Transformation into standard form)

$$\min_{x_1, x_2} \; 10x_1 + 7x_2$$

$$\text{s.t.} \quad 20x_1 + 20x_2 \geq 60$$
$$15x_1 + 3x_2 \geq 15$$
$$x_1 \geq 0$$
$$x_2 \text{ free}$$

$\rightsquigarrow$

First handle the free variable:
Introduce $x_2^+, x_2^- \geq 0$ such that

$$x_2 = x_2^+ - x_2^-$$

After redefining x_2, we can transform each inequality constraint by adding a slack variable. For instance, the first inequality can be reformulated as follows:

$$
\begin{aligned}
& 20x_1 + 20x_2^+ - 20x_2^- \geq 60 \\
\Leftrightarrow\ & -20x_1 - 20x_2^+ + 20x_2^- \leq -60 \\
\Leftrightarrow\ & 20x_1 + 20x_2^+ - 20x_2^- - s_1 = 60, \quad s_1 \geq 0 \\
\Leftrightarrow\ & -20x_1 - 20x_2^+ + 20x_2^- + s_1 = -60, \quad s_1 \geq 0
\end{aligned}
$$

Second inequality analogously.

Transformed LP:

$$
\begin{aligned}
\max_{x_1, x_2^+, x_2^-, s_1, s_2} \quad & -10x_1 - 7(x_2^+ - x_2^-) \\
\text{s.t.} \quad & -20x_1 - 20x_2^+ + 20x_2^- + s_1 = -60 \\
& -15x_1 - 3x_2^+ + 3x_2^- + s_2 = -15 \\
& x_1, x_2^+, x_2^-, s_1, s_2 \geq 0
\end{aligned}
$$

Matrix-vector form: define

$$
\tilde{x} = \begin{pmatrix} x_1 \\ x_2^+ \\ x_2^- \\ s_1 \\ s_2 \end{pmatrix} = \begin{pmatrix} \tilde{x}_1 \\ \tilde{x}_2 \\ \tilde{x}_3 \\ \tilde{x}_4 \\ \tilde{x}_5 \end{pmatrix}, \quad \tilde{c} = \begin{pmatrix} -10 \\ -7 \\ 7 \\ 0 \\ 0 \end{pmatrix}, \quad \tilde{b} = \begin{pmatrix} -60 \\ -15 \end{pmatrix}, \quad \tilde{A} = \begin{pmatrix} -20 & -20 & 20 & 1 & 0 \\ -15 & -3 & 3 & 0 & 1 \end{pmatrix}
$$

Transformed problem has the (standard!) form:

$$
\max_{\tilde{x}} \quad \tilde{c}^T \tilde{x} \quad \text{s.t.} \quad \tilde{A}\tilde{x} = \tilde{b}, \ \tilde{x} \geq 0.
$$

Original variables recovered from $\tilde{x}$ using $x_1 = \tilde{x}_1$, $x_2 = \tilde{x}_2 - \tilde{x}_3 = x_2^+ - x_2^-$.

1.4.2 Properties of LP

Definition 1.15

Consider an LP in standard form.

- A vector $x \in \mathbb{R}^n$ that satisfies all constraints is called **feasible**; otherwise, it is **infeasible**
- The set $C := \{ x \in \mathbb{R}^n \mid Ax = b,\ x \geq 0 \}$ is called the **feasible set**
- The LP is called **feasible** if $C \neq \emptyset$, otherwise **infeasible**
- The **optimal value** of the LP is

$$\nu := \sup \{ c^T x \mid x \in C \}$$

where $\nu \in \mathbb{R} \cup \{-\infty, +\infty\}$. We use the convention $\sup \emptyset = -\infty$.

$\Rightarrow$ An infeasible LP has the "optimal value" $\nu = -\infty$.

- A feasible LP is called **unbounded**, if for every $M \in \mathbb{R}$, there exists a feasible vector $x \in C$ such that

$$c^T x > M.$$

In this case, the "optimal value" is $\nu = +\infty$.

- A feasible LP is called **bounded** if there exists $M \in \mathbb{R}$ such that

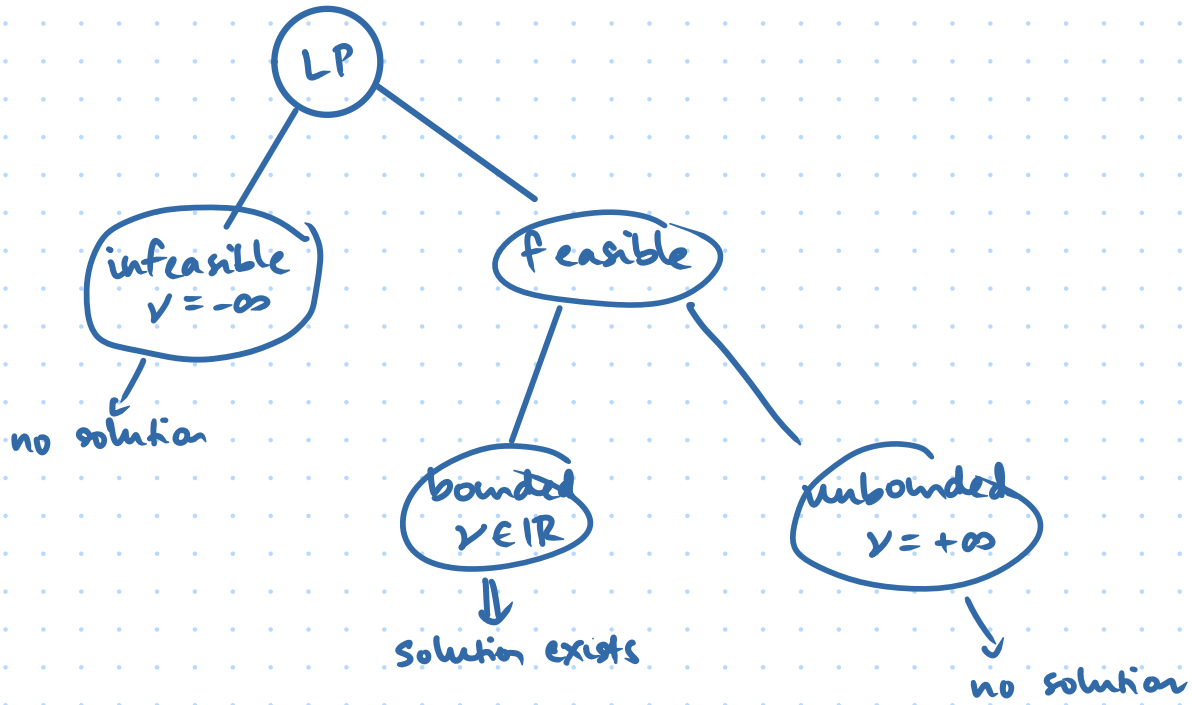
$$c^T x \leq M \qquad \text{for every } x \in C$$

- A feasible point $x^* \in C$ is called an **optimal solution** or a **maximizer** if

$$c^T x^* \geq c^T x \quad \text{for every } x \in C$$

In summary, we have one of the following cases:

1) LP is infeasible, $\nu = -\infty$
2) LP is feasible & unbounded, $\nu = +\infty$
3) LP is feasible & bounded, $\nu < \infty$.



Example 1.16

Consider the LP

$$\begin{aligned} \max \quad & x_1 \\ \text{s.t.} \quad & x_1 \leq 2 \\ & x_1 \geq 3 \end{aligned}$$

$\rightsquigarrow \quad C = \{x_1 \in \mathbb{R} \mid x_1 \leq 2, x_1 \geq 3\} = \emptyset$

$\Rightarrow$ LP infeasible with $\nu = -\infty$

Example 1.17

Consider

$$\begin{aligned} \max \quad & x_1 \\ \text{s.t.} \quad & x_1 - x_2 \leq 1 \\ & x_1, x_2 \geq 0 \end{aligned}$$

$\rightsquigarrow$ LP is feasible, e.g. with $x = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

Consider the vector

$x(t) = \begin{pmatrix} t \\ t \end{pmatrix}$ is also feasible, and the objective value is t.

LP is unbounded for $t \to +\infty$, $\nu = +\infty$, and no solution exists.